

Anuj Sethia

Quantum Photonics Engineer

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HIGHLIGHTS

Competences: Quantum Optics, Integrated Photonics, Atomic Physics, Nano-Fabrication, Cryogenics

Skills: Python, Matlab, Lumerical, Comsol, IPKISS, Virtuoso, Solidworks, Pennylane, QuTip, Qiskit

EDUCATION

PhD in Physics (Fastracked from MSc) Aug 2021 - Dec 2026
University of Calgary, Calgary, Canada

BTech in Mechanical Engineering with Minor in Physics Jul 2016 - Aug 2020
Indian Institute of Technology Bombay, Mumbai, India

PROFESSIONAL EXPERIENCES

Junior Application Engineer	Luceda Photonics, Remote	Sep 2026 - Dec 2026
Research Assistant	QNu Labs, Bangalore, India	Jun 2020 - July 2021

KEY PROJECTS

Photonics PDK Development Sep 2025 - Dec 2026
Luceda Photonics | Remote

- Delivered a Si and SiN Process Development Kit in IPKISS for a commercial photonics foundry
- Built parameterized component models and DRC decks to validate layout compliance and simulation

TFLN based Quantum Memory Aug 2025 - Dec 2026
Advisor: Prof. Daniel Oblak | University of Calgary

- Developed a cryogenic setup for reliable low loss fiber-to-chip coupling of integrated photonic chips
- Demonstrated record-long on-demand storage of telecom-band photons in Erbium-doped thin-film lithium niobate (TFLN) waveguides
- Designed and demonstrated on-chip cavity enhancement to improve quantum memory efficiency

TFLN Nano-Fabrication Jun 2024 - Dec 2026
Advisor: Prof. Daniel Oblak | University of Alberta

- Established a multi-step nano-fabrication process for TFLN photonic devices by optimizing steps, including electron-beam lithography and reactive-ion etching.
- Developed a process design kit (PDK) enabling rapid design and layout of photonic components
- Achieved low propagation losses of 0.25 dB/cm with a scalable process for quantum photonics

Millisecond-Scale Telecom Quantum Memory Jan 2025 - Dec 2025
Advisor: Prof. Daniel Oblak | University of Calgary

- Engineered an alignment-free interface at cryogenic temperature for scalable quantum repeaters
- Demonstrated a telecom-band erbium-ion quantum memory with a record 1 ms storage time
- Increased memory capacity using spectral-temporal multiplexing for scalable quantum networking

Silicon Quantum Photonics Jan 2023 - Jun 2024
Advisor: Prof. Daniel Oblak | University of Calgary

- Engineered silicon photonic device designs using Lumerical for enhanced nonlinear optics
- Developed a Python-based die testing platform for data acquisition and rapid device testing
- Demonstrated on-chip generation of entangled photon pairs for integrated quantum communication

Measurement Device Independent Quantum Key Distribution

Sep 2021 - Jun 2023

Advisor: Prof. Daniel Oblak | University of Calgary

- Collaborated with a cross-functional team to develop a minimum viable product for MDI-QKD
- Demonstrated high-fidelity Bell-State Measurements (BSM) between independent time-bin qubit
- Patented feedback control mechanisms enabling reliable long-distance quantum communication

Quantum Key Distribution

Jun 2020 - Jul 2021

QNu Labs | Bangalore, India

- Developed a MATLAB-based simulation framework to model the performance of QKD systems
- Evaluated the information-theoretic security of the QKD protocol against advanced eavesdropping
- Extended secure key exchange to over 100 km (2.5x improvement) through parameter optimization
- Led ideation and early-stage development of a quantum-secure video conferencing service

AWARDS & SCHOLARSHIPS

Cannon Lachapelle Award	Hunter Hub, UCalgary	2026
Optics and Photonics Scholarship	SPIE	2026
Evolve to Innovate	Hunter Hub, UCalgary	2026
Start Something Weekend	Hunter Hub, UCalgary	2026
MicroFAB Access	CMC Microsystems	2025
Graduate Student Scholarship	Alberta Innovates	2023-25
Best Poster Presentation	IEEE Photonics	2023
Best PIC Design Award	CMC Microsystems	2023
Alberta Graduate Excellence Scholarship	Province of Alberta	2022
Faculty of Graduate Studies Scholarship	UCalgary	2022
Entrance Scholarship	Dept. of Physics & Astro.	2021
Undergraduate Research Award	IIT Bombay	2020

PROFESSIONAL DEVELOPMENT

- **Reviewer** @ PRA, PRL, PRX Quantum, Nature Physics Jul 2024 - Present
- **Completed** Evolve to Innovate program for translating research into impact Jan 2026 - May 2026
- **Member** @ Research Innovation & Scholarship Committee, UCalgary Sep 2025 - Apr 2026
- **Member** @ Quantum Horizons Alberta Director Hiring Committee, UCalgary Sep 2025 - Apr 2026
- **Participant** @ GenQ Quantum Hackathon Oct 2024
- **Finalist** @ Three Minute Thesis Competition Apr 2024
- **Organizer** @ Quanta Create Symposium 2023, UAlberta, Edmonton Oct 2022 - Mar 2023
- **Secretary & Treasurer** @ Calgary Optics and Photonics Students Society Jan 2022 - Jan 2023
- **Mentor** @ Student Mentorship Program, IIT Bombay Aug 2019 - Jul 2020

PATENTS & PUBLICATIONS

1. **Sethia, A.**, Kamel, N. G., & Oblak, D. (2025). On-Demand millisecond storage of Spectro-Temporal multimode telecom photons. Phys. Rev. Lett.
2. Singh, A., **Sethia, A.**, Esmaeilifar, L., Valivarthi, R., Sinclair, N., Spiropulu, M., & Oblak, D. (2026). Photonic quantum information with time-bins: Principles and applications. Phys. Reports, 1193, 1–79.
3. Smith, J. D.; Oblak, D.; **Sethia, A.**; Chenini, H.; Singh, A.; Lucio Martinez, I. (2023) Apparatus and Methods for Ongoing Feedback during Quantum Key Distribution. US Patent [Pending]
4. **Sethia, A.**, & Banerjee, A. (2022). A MATLAB based modeling and simulation package for DPS-QKD. Journal of Modern Optics, 69(7), 392–402.
5. Banerjee, A., **Sethia, A.**, Mogiligidra, V., Krishnan, R. K., AR, M., Rajamani, S., & Shenoy, V. (2021). Experimental demonstration of high-entropy time of arrival based optical QRNG qualifying stringent statistical tests. arXiv preprint

